

A placebo-controlled randomized cross-over lovastatin trial in neurofibromatosis type 1 reveals Ras/MAPK modulation of dynamic emotional face processing

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Abstract

Neurofibromatosis type 1 (NF1) is a neurodevelopmental disorder characterized by cognitive and socioemotional processing deficits, associated with dysregulation of the Ras/MAPK pathway and an impact on neural inhibition and the GABA system. This study examined the effects of lovastatin, a Ras/MAPK pathway modulator with a direct influence on excitatory/inhibitory (E/I) balance, on neural responses during an emotional recognition task using a randomized, triple-blind, placebo-controlled crossover design. Electroencephalography (EEG) oscillatory responses, was recorded pre- and post-intervention, focusing on parietal theta rhythms which were previously shown to be related to inhibition and index dynamic face emotion recognition. In this within subject cross-over design, fifteen NF1 individuals were pseudorandomly allocated to placebo (3 doses of 60 mg) and lovastatin (3 doses of 60 mg) conditions. Data were collected at pre-intervention, post-placebo and post-lovastatin intake and analyzed using event-related spectral perturbations (ERSP) computed with Morlet wavelets. Lovastatin selectively modulated theta band power (5–6 Hz) during emotional recognition, as reflected by a significant intervention $\times$ task interaction and a significant within-session reduction in centroparietal theta oscillations following lovastatin. No significant effects were observed in alpha (8–11 Hz) or beta (13–20 Hz) bands, nor in theta power during visual control conditions. Given the established role of theta oscillations in face emotion processing, the observed reduction suggests lovastatin influences task-specific oscillatory dynamics. As lovastatin these findings provide a mechanistic link between theta oscillations, emotion recognition, and inhibitory regulation. This novel neurophysiological effect identifies a potential treatment target for NF1 and related neurodevelopmental disorders, including autism (NIH Trial Registration #NCT03826940).

Keywords Neurofibromatosis type1 · Lovastatin · Theta brain oscillations · Excitatory/inhibitory balance

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Introduction

Neurofibromatosis type 1 (NF1) is a multisystem neurodevelopmental disorder caused by mutations in the *NF1* gene, with a global prevalence of approximately 1 in 3000 individuals¹. It is associated with diverse clinical features, including café-au-lait macules, benign and plexiform neurofibromas, skeletal abnormalities, and, in some cases, malignant peripheral nerve sheath tumors^{2,3}.

In addition to its physical manifestations, NF1 significantly impacts cognitive, social, and emotional domains. Up to 50–70% of individuals with NF1 experience deficits in working memory, emotional and social reciprocity, and communicative abilities, often accompanied by repetitive behaviors⁴. These challenges contribute to difficulties in recognizing and interpreting emotions and broader impairments in social cognition, including the theory of mind. Such deficits may explain the difficulties in social interaction and poorer social outcomes observed in children with NF1^{5–7}. The frequent co-occurrence of neurodevelopmental conditions like Autism Spectrum Disorder (ASD) and Attention Deficit Hyperactivity Disorder (ADHD), along with an increased risk of intellectual disability and epilepsy, underscores the complexity of NF1 and highlights the need for multidisciplinary management approaches. Addressing these challenges is essential to improve the quality of life and tailor interventions for individuals with NF1^{8,9}.

It has been established that the molecular basis of neural phenotypes in NF1 is based on a neurotransmission imbalance of the γ -aminobutyric acid (GABA) system, the brain's primary inhibitory neurotransmitter^{9,10}. Mutations in the *NF1* gene, which encodes neurofibromin, disrupt the Ras/MAPK signaling pathway, leading to GABAergic dysfunction and increased synaptic inhibition, impaired neural plasticity and cognitive deficits¹¹.

Evidence from both animal models and human studies supports the presence of an excitatory/inhibitory (E/I) imbalance in NF1, characterized by reduced GABA levels and altered GABAergic signaling^{8,9,12,13}. Reviews and cross-species synthesis further argue that NF1-related cognitive dysfunction reflects disrupted inhibitory tone, with neurofibromin acting preferentially in inhibitory circuits and altering both pre- and postsynaptic GABAergic regulation^{14–16}. These neurochemical alterations in individuals with NF1 are directly related with contrast processing, which is a direct consequence of excessive inhibition and atypical patterns of activation of the default mode network (DMN)¹⁷, reflected in impaired task-related deactivation, which have been associated with attentional deficits¹⁷. As GABAergic neurotransmission is a key regulator of DMN connectivity and deactivation, these findings provide a mechanistic link between GABAergic dysfunction and large-scale network abnormalities in NF1^{16,17}.

Both evidence from NF1 mouse models and human studies are consistent with E/I unbalance^{9–14}. Magnetic resonance spectroscopy studies have shown reduced GABA levels in NF1 patients across multiple brain regions and changes in GABA receptor binding (9,10). These disruptions are particularly evident in brain regions like the visual cortex, medial frontal cortex, and thalamus, supporting the idea of a neurodevelopmental synaptopathy underlying the cognitive and behavioral challenges in NF1²⁰.

Pharmacological interventions targeting the Ras/MAPK pathway have shown potential in addressing these GABAergic imbalances. Lovastatin, a 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase inhibitor, has been extensively studied for its role in modulating Ras activity. By inhibiting p21Ras isoprenylation, lovastatin reduces Ras hyperactivation, which has been implicated in the cognitive deficits observed in NF1²¹. Studies in the *Nf1*^{+/-} mouse model have demonstrated that lovastatin normalizes synaptic plasticity and restores E/I balance, improving spatial learning, memory, and attention^{21,23,24}.

Human clinical trials evaluating lovastatin have reported promising results. For instance, treatment with lovastatin was associated with significant improvements in memory and attention in children with NF1, as well as normalization of functional connectivity within the default mode network. These findings suggest that lovastatin not only mitigates Ras/MAPK-driven synaptic dysfunction but also modulates GABAergic neurotransmission, addressing key neurochemical deficits in NF1^{10,21,25,26}. While initial results are encouraging, further large-scale,

randomized controlled trials are needed to confirm the efficacy of lovastatin as a therapeutic option for cognitive deficits in NF1^{27,28}.

Facial emotion recognition has well-established EEG correlates that help contextualize why this task is informative in NF1 in relation to GABA deficits and changes in brain oscillations. Early visual encoding is reflected in the P100, structural face processing in the N170, and later evaluative stages in the P250, while theta oscillations support long-range communication and emotional appraisal^{29,30}. Neurodevelopmental conditions marked by E/I imbalance, such as ASD, often show reduced N170 amplitude, atypical P250 modulation, and altered theta dynamics. These markers clarify what readers should expect in NF1: deviations in ERP components or oscillatory patterns may reflect the same inhibitory dysfunction and network disorganization described in the disorder, providing a direct electrophysiological window into socio-emotional processing³¹.

Prior studies were able to show a direct relation between GABAergic dysfunction in individuals with NF1, and cognitive deficits stemming from disruptions in the brain's excitatory/inhibitory (E/I) balance^{9,32}. This imbalance may be reflected in altered neural oscillations, which play a critical role in various cognitive and emotional processes, including theta oscillations in facial expression recognition³³. The impact of GABA signaling in shaping neural oscillations is quite relevant in health and disease. EEG studies have shown that individuals with NF1 exhibit atypical oscillatory patterns in line with other neuropsychiatric disorders^{34–40}, particularly within frequency bands associated with emotional perception and processing^{33,41,42}. Regarding EEG correlates of face processing and GABA our prior studies by Sousa et al. do indeed confirm a link between EEG signals, social cognition and GABA levels^{41,42}.

In the alpha band, NF1 participants exhibit a reduced peak alpha frequency during resting-state, a marker closely associated with neural efficiency and brain maturation³⁴. Lower alpha frequency may reflect delayed or atypical neurodevelopmental trajectories. Moreover, alpha oscillations are typically linked to inhibitory control mechanisms, suppressing irrelevant information and supporting top-down attentional regulation³⁷. NF1 individuals display paradoxically enhanced alpha and beta desynchronization in response to peripheral visual targets³⁵. This abnormal modulation suggests the presence of compensatory mechanisms and points to disrupted context-dependent regulation of attentional processes^{34,38}.

Theta activity, which is critically involved in long-range cortical communication and emotional processing³⁹, also appears altered in NF1. During working memory tasks, NF1 individuals demonstrate increased theta power and frontoparietal phase coherence, patterns that are generally interpreted as supporting cognitive engagement. However, these differences diminish when accounting for resting-state baseline activity, indicating that task-related effects may reflect pre-existing resting-state abnormalities^{34,35,40}.

Importantly, we have recently shown that parietal theta oscillations index emotion faces processing, both regarding visual processing and mental imagery, as validated by machine learning approaches³³. This study showed that in related neurodevelopmental conditions such as ASD reveal there is theta activity during emotional imagery, particularly in the right precuneus, a region implicated in mental imagery and self-referential processing. Such findings suggest that theta oscillations may be a key marker of atypical emotional processing in neurodevelopmental disorders³³.

These alterations task-related oscillatory dynamics point to a global disruption in the temporal coordination of neural activity in NF1, particularly at a frequency bands related to emotional processing motivated the current study. In this context, lovastatin has been investigated as a potential pharmacological intervention to restore E/I balance by modulating GABAergic activity and oscillatory activity with a focus on the theta band, while investigating other frequency bands. Based on prior evidence linking theta oscillations to facial emotion expression recognition³³, we hypothesized that lovastatin would modulate theta-band activity during facial emotion recognition in NF1 in this specific ranges. This hypothesis-driven approach aims to clarify how pharmacological modulation of inhibitory tone influences the neural dynamics underlying emotional processing in NF1, while also investigating other control frequency bands.

Methods

Participants

A total of 15 NF1 patients, aged between 26 and 55 years ($\text{mean} \pm \text{SE} = 39.60 \pm 2.08$ years), were enrolled in this study carried out at the CIBIT/ICNAS facilities (University of Coimbra, Portugal). All patients met the National Institute of Health (NIH, Consensus Development Conference 1988) diagnostic criteria for NF1. Exclusion criteria were as follows: other neurological or psychiatric disorders, abnormal liver function, history of traumatic brain injury, epilepsy, substance abuse, severe learning disabilities (WAIS-III, $\text{IQ} < 70$), previous or current statin use and usage of medication with potential interaction with Lovastatin. All participants received an abbreviated form of the Portuguese-adapted version of the Wechsler Adult Intelligence Scale.

Study procedures were reviewed and approved by the Ethics Committees of the Faculty of Medicine of the University of Coimbra (approval number CE-007/2017) and performed following the Declaration of Helsinki and all relevant guidelines. All participants provided written informed consent before their inclusion.

Study design and procedures

A randomized, triple-blind, placebo-controlled Phase 2a Clinical Trial, first registered at clinicaltrials.gov on 01/02/2019, with registration number #NCT03826940 (ASD/NF1inhib), was conducted to investigate the physiological and neurochemical effects of lovastatin in adults with NF1. Following a crossover design, all subjects were allocated to both real and placebo interventions. Sample size estimation is based on Ref²¹.

In total, three visits were conducted, involving the acquisition of EEG data (Fig. 1). After the initial session, pre-intervention session (visit 1 After the initial session (pre-intervention session; visit 1), participants were randomly assigned into two groups, comprising seven and eight participants, for the three-day intervention period: about half of the participants randomly took lovastatin, while the other half took a placebo for three days. Then they returned, on the immediate next day, for the second visit. Subsequently, a minimum of four weeks (range: 33–110 days, $\text{mean} = 53.27 \pm 24.15$ days) washout interval occurred. Three days before the third visit, participants resumed taking either a lovastatin pill or a placebo, following the reverse order from the second visit. The participants performed the same cognitive task in all EEG sessions/visits.

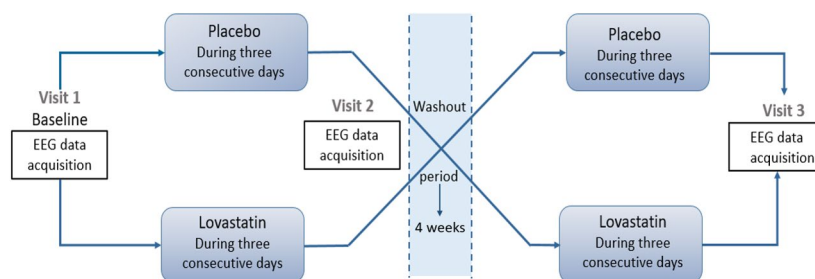


Fig. 1 Flowchart of study design and procedures were performed in each visit. In total, three visits were conducted, involving the acquisition of EEG data in all of them. After the initial session (visit 1), participants were randomly divided into two groups: half of the participants took lovastatin, while the other half took a placebo for three days. Then they returned for a second visit, after which a four-week washout interval occurred. Three days before the third visit, participants resumed taking either a lovastatin pill or a placebo, following the reverse order from the second visit. The participants performed the same cognitive task in all EEG sessions/visits.

Medication protocol and randomization

The intake of lovastatin or placebo was randomized for each participant using a computer-generated 1:1 permutation-block randomization list. Sequentially numbered containers ensured proper allocation. Randomization procedures, including participant assignment and drug distribution, were managed by the Clinical Research Unit of the University of Coimbra. Both patients and investigators were blinded to treatment allocation, and the authors remained unaware of group assignments until the completion of data analysis.

Participants received 60 mg per day (three 20 mg pills) of either lovastatin or placebo, identical in color, size, and shape (to ensure blinding), for three consecutive nights. Compliance and adverse events were monitored through phone contact (once a day, at the end of the day). Participants were unable to distinguish between the lovastatin and placebo conditions and reported no symptoms attributable to lovastatin intake.

CONSORT checklist and flow diagram are presented as supplementary materials.

Paradigm

Dynamic facial expressions in a virtual avatar (happy, sad, and neutral facial expressions) (Fig. 2) were presented to the participants during the EEG recording (as in³³). Each trial is composed by a neutral expression (1 s), followed by a morphing period of 250 ms going from the neutral to a sad or happy expression, which then remains for 1 s, and a final period where the avatar morphs back to the neutral expression (250 ms), which then remains for 1 s. At the end of the trial, participants should indicate whether the expression seen was happy or sad by pressing button 1 or 2, respectively. The response period lasted until a button press occurred or for 1 s (maximum) in cases where no answer was registered. Each trial is always preceded by a black screen baseline time of 1s. This experiment consisted of 3 runs of 4.17 min each, with 50 trials per run – 25 for happy and 25 for sad conditions, and short breaks in between. A total of 75 trials per condition were recorded, and the full length of the task was around 12.5 min.

The visual stimulation task was developed in *WorldViz Vizard 5 VR Toolkit* (development edition) using the male002 virtual avatar from the *Complete Characters HD pack* and its FE poses. The virtual avatar measured 700 × 530 pixels, and the emotional expressions increased by 6.67% per frame.

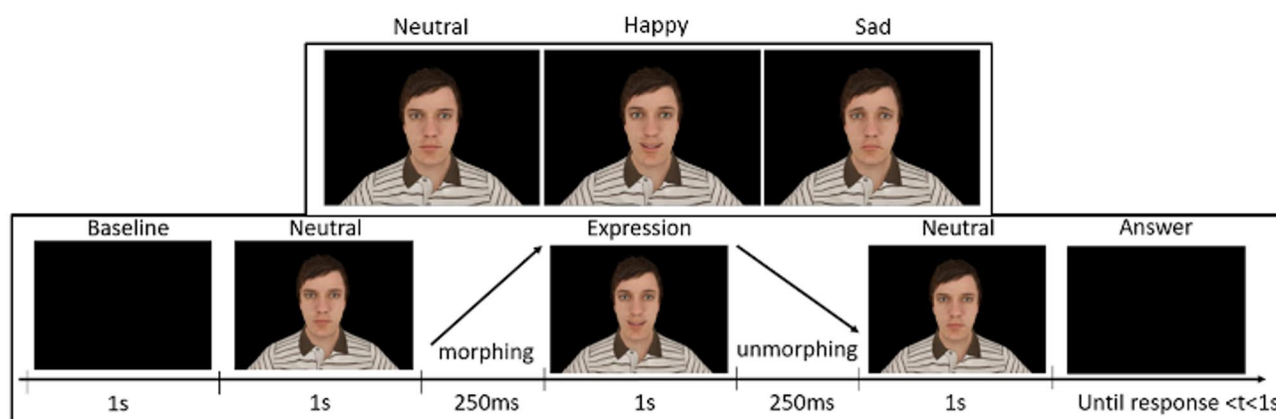


Fig. 2 Experimental task overview: The paradigm incorporates three facial expressions: neutral, happy, and sad. Each trial begins with a 1-second presentation of a virtual avatar with a neutral expression, followed by a 250-millisecond morphing transition. The avatar then displays either a happy or sad expression for 1 s. The trial ends with the avatar morphing back to a neutral expression for an additional 1 s. A black screen baseline of 1 s precedes each stimulus, followed by a response phase.

EEG acquisition and processing

The experiment was carried out using a 22-inch LCD monitor with a 60 Hz refresh rate and a resolution of 1680×1050 pixels. The experimental paradigm was implemented in Matlab[®] (MathWorks, version R2017a). Participants were seated at an approximate distance of 60 cm from the display.

EEG data were acquired with an actiCAP system comprising 64 active Ag/AgCl electrodes arranged according to the international 10–10 system and connected directly to a Brain Products actiCHamp amplifier. Signals were sampled at 1000 Hz. The ground electrode was positioned at AFz, while the reference electrode was placed at FCz. Electrode impedances were maintained below 15 k Ω . EEG recordings were collected using BrainVision Recorder software (Brain Products, version 1.20.0801).

We used Matlab[®] (Mathworks, version 2023a) and the EEGLAB toolbox 2019.0 for the EEG.

processing and analysis. EEG data were downsampled to 500 Hz and filtered with a finite response bandpass filter with lower and higher cut-off frequencies set to 1 and 100 Hz, respectively. Noisy channels were then removed by visual inspection and interpolated. Afterward, the data were re-referenced to the common average reference. After that, a stop-band filter at 50 Hz. Noisy epochs were identified and rejected by visual inspection, resulting in an average rejection rate of 10.6% across all participants. Finally, Independent Component Analysis (ICA)³⁹ was performed, and components corresponding to visual artifacts (e.g., eye blinks and eye movements) were identified by visual inspection and removed.

Data analysis

Event-related spectral perturbation (ERSP) was computed using Morlet wavelets with incremental cycles (2 cycles at 0.5 Hz, up to 5 at 100 Hz), resulting in 200 time points (from -200 pre-stimulus to 1500 ms after stimulus onset) and 100 frequency points. ERSP analyses were performed using comparing oscillatory power across conditions including the neutral face baseline. The analysis focused on the first 1500 ms of the task, as we aim to investigate the neural correlates of emotion recognition.

Event-related spectral Perturbations were plotted using the averaged topography for two distinct temporal windows. The first window spanned 0–270 ms, corresponding to the visual attention phase during which participants observed a static face with a neutral expression. The second window covered 1000–1270 ms, corresponding to the emotional recognition phase, beginning with morphing the face toward an emotional expression and extending up to 70 ms after the face assumed a static expression (happy or sad)^{41,42}.

Channel selection for subsequent analysis was guided by previous findings linking centro-parietal theta activity to emotional processing³³ and by evidence associating parieto-occipital alpha and beta desynchronization with attentional engagement and visual modulation during facial emotion recognition. So, two electrode clusters were selected: (CP1, CPz, CP2)^{33,43} and (PO7, POz, PO8)^{40,44}. The first cluster was used for the analysis of theta-band activity (5–6 Hz), while the second cluster was employed to calculate alpha (8–11 Hz) and beta-band activity (13–20 Hz).

For each participant and condition, ERSP values were averaged across all time and frequency bins within the selected temporal and frequency ranges. These values were then averaged across the electrodes within each predefined cluster, yielding a single mean ERSP value per cluster and condition for statistical analysis.

Data were analysed separately for theta, alpha, and beta frequency bands, considering a 3×2 factorial design with intervention (pre-intervention, post-placebo, post-lovastatin) and 2 task periods (initial visual neutral face presentation and emotional recognition) as within-subject factors, resulting in 6 category-specific conditions. Repeated-measures ANOVAs were performed on the predefined time-frequency windows for the three frequency bands. We assessed sphericity using Mauchly's test; when necessary, Greenhouse-Geisser corrections were applied. For post-hoc comparisons, Bonferroni correction was used. The adopted significance level was $\alpha = 0.05$. Reported effect sizes are partial η^2 .

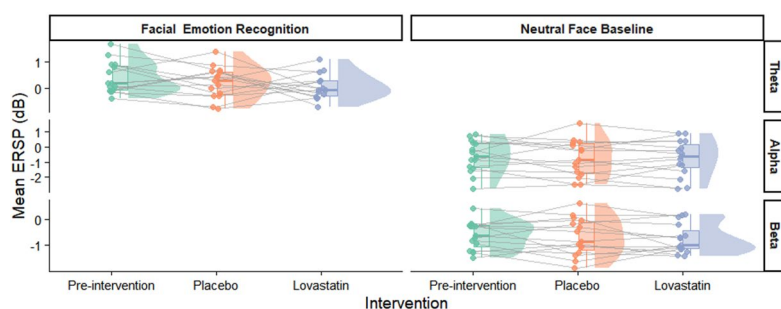


Fig. 3 The figure illustrates the distribution of mean event-related spectral perturbation (ERSP) values across the three intervention phases (pre-intervention, placebo, and lovastatin). Theta-band activity (5–6 Hz) was quantified from the centroparietal electrode cluster (CP1, CPz, CP2) during the facial emotion recognition period (1000–1270 ms). Alpha (8–11 Hz) and low-beta (13–20 Hz) activity were quantified from the parieto-occipital electrode cluster (PO7, POz, PO8) during the baseline control period (0–270 ms), corresponding to the presentation of the neutral face. Violin plots represent the distribution of individual ERSP values, boxplots indicate the median and interquartile range, and dots represent individual participants. Individual participant lines are included to depict within-subject trajectories across intervention phases, allowing visualization of the tendency and direction of change for each participant.

Results

To investigate the effects of lovastatin on NF1, we analysed the power variation of the theta band (the main frequency of interest, given the prior work of Simões, 2018³³ showing a direct link with facial emotion recognition), alpha, and low-beta neural oscillations during a cognitive task focused on two moments: one centered on the initial response to a neutral face and the other on the recognition of emotional facial expressions.

First parietal alpha (8–11 Hz) and low-beta (13–20 Hz) oscillations, known to be crucial for the processing of visual information, were analysed for the temporal interval 0 to 270ms corresponding to the window of presentation of a static neutral face. As illustrated in the t Fig. 3, there are no significant differences as tested based on an ANOVA with Bonferroni correction test between the intervention phases and the pre-intervention condition. The alpha and beta topographies show a highly homogeneous scalp distribution during this early perceptual window, with no focal increases or decreases in parietal, occipital or frontal sites across the three phases. For the alpha band (Fig. 3), subtraction maps demonstrated a consistent posterior (parieto-occipital) increase, reflecting greater alpha power at baseline relative to the emotional condition, a pattern typically associated with alpha suppression during cognitively demanding tasks. This effect appeared across all groups, without robust differences between interventions. In the beta band, subtraction maps were uniformly flat, indicating no relevant differences between baseline and emotional recognition across interventions. These results indicate that lovastatin did not alter the early oscillatory dynamics typically linked to sensory encoding and early visual processing of neutral faces.

In the second temporal interval (1000–1270 ms), corresponding to the morphing of a static neutral face into an emotional expression (happy or sad), theta band power (5–6 Hz) reveal a pronounced reduction of theta power over parietal/posterior sites, most clearly in the lovastatin group (Fig. 3.) The subtraction maps all groups showed a widespread reduction of power during the emotional recognition interval compared to baseline, with this decrease being most pronounced in the Lovastatin group, suggesting a pharmacological modulation of late theta activity. The repeated-measures ANOVA revealed a significant interaction between intervention and task, $F^{2,28}=4.043$, $p=0.029$, partial $\eta^2=0.224$. Indicating that the change in theta power between the neutral-face and emotional-recognition periods differed across intervention sessions. Post hoc comparisons showed that only the lovastatin session exhibited a significant within-session reduction in theta power from the neutral-face to the emotional-recognition period (mean difference = 0.504, SE = 0.165, $p=0.009$), whereas the corresponding within-session comparisons were not significant for the pre-intervention ($p=0.704$) or placebo ($p=0.163$) sessions. Thus, the primary evidence supporting a differential intervention effect is the significant intervention $\times$ task interaction.

The results suggest that the experimental manipulation Baseline vs. emotional recognition modulates theta power in an intervention-dependent manner, as reflected by the significant intervention $\times$ task interaction. The significant within-session reduction observed following lovastatin, together with the absence of significant reductions in the other intervention phases, is also consistent with the hypothesis that lovastatin influences the task-related modulation of theta oscillations.

Behavioral data

Behavioral performance was monitored throughout the experiment. In the post-placebo session, there were 5 incorrect responses out of 136 with a mean reaction time of 453 ms, whereas in the post-lovastatin session, there were 3 incorrect responses out of 130, with a mean reaction time of 451 ms. Overall, behavioral performance remained comparable across intervention sessions and did not suggest differences in task engagement that could account for the observed electrophysiological findings.

Discussion

This study investigated the effects of lovastatin on neural oscillations, with a focus on theta oscillations related to facial expression processing in individuals with NF1, using an emotional recognition task. The findings provide novel insights into the potential of lovastatin to modulate face emotional processing associated with NF1 by influencing theta band activity.

Our paradigm choice was based on prior studies of face processing have confirmed a link between EEG signals, social cognition and GABA levels^{41,42}. The focus on theta rhythms as a therapeutic target was based on a previous study linking directly facial expression recognition with theta rhythms³³.

During presentation of a neutral face, lovastatin did not induce significant alpha and beta band activity alterations. These results are in line with previous research showing limited effects of lovastatin on basic sensory processing^{21,45} and suggest that oscillatory mechanisms, at these frequency bands remain stable across intervention phases. Parieto-occipital alpha desynchronization, which reflects top-down inhibition of task-irrelevant information and efficient allocation of attentional resources, was consistently observed while visualizing a neutral face. Similarly, beta desynchronization, often linked to the dynamic modulation of visual input and sensorimotor integration, did not vary significantly^{46,47}. These findings reinforce the interpretation that lovastatin's neuromodulatory effects may be specific to the emotional component of face processing rather than reflecting a generalized modulation of neural activity. This interpretation is also in line with previous studies reporting limited evidence for lovastatin-related improvements in other cognitive domains, for example, Payne²² reported that lovastatin did not affect the primary outcomes of visuospatial learning or attention, and Bearden⁴⁸ similarly found no improvements in measures of attention following lovastatin administration.

Lovastatin administration was associated with a significant within-session reduction in centroparietal theta power between the neutral-face baseline and the emotional-recognition period. This reduction emerged specifically during the 1000–1270 ms time window of our paradigm, coinciding with the dynamic unfolding and recognition of the emotional expression. In individuals with NF1 and ASD, elevated theta activity has been interpreted as a marker of excitatory/inhibitory imbalance and inefficient neural communication^{32,33}. The observed reduction in theta power is consistent with the hypothesis that lovastatin modulates the altered GABAergic signaling described in NF1. Rather than reflecting a generalized suppression of neural activity, the reduction in theta oscillations may normalize E/I dynamics, potentially improving the efficiency of neural circuits that support restoration of excitatory/inhibitory balance.

However, these findings should be interpreted with caution. As this placebo controlled study did not include a healthy control group, we cannot conclude that the observed modulation represents necessarily a normalization of excitatory/inhibitory dynamics or a restoration of typical neural function. Likewise, although the reduction in

theta power may reflect changes in attentional resource allocation or greater automatization of emotional processing, these remain candidate interpretations. Importantly, behavioural performance, including accuracy, missed responses, premature responses, and reaction times, was comparable across intervention sessions, arguing against a generalized reduction in task engagement, although this possibility cannot be entirely ruled out.

Mechanistically, these findings are consistent with the proposed biological action of lovastatin in NF1. Neurofibromin deficiency leads to hyperactivation of the Ras signaling pathway, which has been associated with increased GABA release and disrupted excitatory/inhibitory (E/I) balance in cortical circuits. Experimental studies in NF1 mouse models have shown that lovastatin reduces Ras activity, normalizes excessive GABAergic neurotransmission, and restores synaptic plasticity. By re-establishing a more balanced inhibitory tone, lovastatin may improve the temporal coordination of neuronal populations involved in socio-emotional processing, thereby explaining the selective modulation of theta oscillations observed during emotional recognition^{24,49}.

This modulation pattern highlights that lovastatin's effects are not globally distributed across the brain but are rather selectively recruited in the context of emotion-driven cognitive demands. By targeting neural oscillations that are dynamically engaged during socio-emotional tasks, lovastatin may restore oscillatory modulations in brain regions implicated in facial emotion processing and known to show altered activity in this population^{10,25}.

Furthermore, these results align with prior studies showing lovastatin-induced improvements in memory and attentional control and enhanced functional connectivity^{22,26}. In this context, the selective theta modulation observed here may reflect a shift toward more efficient information processing, allowing NF1 individuals to engage more adaptively with socially relevant cues.

Despite these promising findings, some limitations must be considered. First, the short intervention period (3 days per condition) and the crossover design do not allow inferences about long-term neural adaptations. While the reduction in theta activity during emotional recognition is promising, because it can be related with excessive theta activity related to facial emotion processing in disorders such as autism, it may reflect only transient neurophysiological changes rather than lasting improvements. Second, the relatively small sample size (in spite of the within subject design) limits the generalizability of the results and may reduce the power to detect subtler effects in other frequency bands or brain regions.

In conclusion, this study provides evidence that lovastatin selectively influenced the modulation of theta oscillations associated with emotional recognition in individuals with NF1, while baseline-related oscillations remain unaffected. These findings are consistent with the hypothesis that lovastatin modulates GABAergic dysfunction associated with NF1, thereby contributing to a more balanced excitatory/inhibitory state in neural circuits involved in socio-emotional processing. Rather than exerting a broad neuromodulatory effect, lovastatin appears to act in function and frequency-specific manner, targeting those neural dynamics related to socioemotional cognition. These results underscore the therapeutic potential of lovastatin for addressing key cognitive-emotional deficits in neurodevelopmental disorders and highlight the importance of considering neural oscillations as sensitive markers of pharmacological intervention effects.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-72672-y>.

Author contributions Diana Costa: Methodology, formal analysis, investigation, data curation, validation, writing - original draft. Teresa Sousa: Methodology, data curation, validation, writing - review and editing. Ana Espírito: data acquisition. Ana Dionísio: participants' follow-up and support, data acquisition. Inês Bernardino: participants' follow-up and support, data acquisition. Marco Simões: Methodology. Ricardo Martins: Design of the work. Miguel Castelo Branco: Conceptualization, methodology, Writing - review and editing, supervision, project administration, funding acquisition.

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Data availability The datasets generated during and analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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